

A Frog 1

atcoder/frog.cpp

```
#include <bits/stdc++.h>
using namespace std;
using ll = long long;

int main() {
    cin.tie(0)->sync_with_stdio(0);

    int n;
    cin >> n;

    vector<ll> h(n), dp(n, 1e18);
    for (ll &x : h)
        cin >> x;

    dp[0] = 0;

    for (int i = 1; i < n; i++) {
        dp[i] = dp[i - 1] + abs(h[i] - h[i - 1]);

        if (i >= 2)
            dp[i] = min(dp[i], dp[i - 2] + abs(h[i] - h[i - 2]));
    }

    cout << dp[n - 1] << '\n';
}
```

B Frog 2

atcoder/frog2.cpp

```
#include <bits/stdc++.h>
using namespace std;
using ll = long long;

int main() {
    cin.tie(0)->sync_with_stdio(0);

    int n, k;
    cin >> n >> k;

    vector<ll> h(n), dp(n, 1e18);
    for (ll &x : h)
        cin >> x;

    dp[0] = 0;

    for (int i = 1; i < n; i++) {
        for (int j = 1; j <= k && i - j >= 0; j++) {
            dp[i] = min(dp[i], dp[i - j] + abs(h[i] - h[i - j]));
        }
    }

    cout << dp[n - 1] << '\n';
}
```

C Vacation

atcoder/vacation.cpp

```
#include <bits/stdc++.h>
using namespace std;
using ll = long long;

int main() {
    cin.tie(0)->sync_with_stdio(0);

    int n;
    cin >> n;

    ll a, b, c;
    cin >> a >> b >> c;

    ll x = a, y = b, z = c;

    for (int i = 1; i < n; i++) {
        cin >> a >> b >> c;

        ll nx = a + max(y, z);
        ll ny = b + max(x, z);
        ll nz = c + max(x, y);

        x = nx;
        y = ny;
    }
```

```
        z = nz;
    }

    cout << max({x, y, z}) << '\n';
}
```

D Knapsack 1

atcoder/knapsack1.cpp

```
#include <bits/stdc++.h>
using namespace std;
using ll = long long;

int main() {
    cin.tie(0)->sync_with_stdio(0);

    int n, W;
    cin >> n >> W;

    vector<ll> dp(W + 1);

    for (int i = 0; i < n; i++) {
        int w, v;
        cin >> w >> v;

        for (int j = W; j >= w; j--)
            dp[j] = max(dp[j], dp[j - w] + v);
    }

    cout << dp[W] << '\n';
}
```

E Knapsack 2

atcoder/knapsack2.cpp

```
#include <bits/stdc++.h>
using namespace std;
using ll = long long;

int main() {
    cin.tie(0)->sync_with_stdio(0);

    int n;
    ll W;
    cin >> n >> W;

    vector<pair<ll, int>> a(n);
    int sum = 0;

    for (auto &[w, v] : a) {
        cin >> w >> v;
        sum += v;
    }

    const ll INF = 4e18;
    vector<ll> dp(sum + 1, INF);
    dp[0] = 0;

    for (auto [w, v] : a) {
        for (int j = sum; j >= v; j--)
            dp[j] = min(dp[j], dp[j - v] + w);
    }

    int ans = 0;
    for (int v = 0; v <= sum; v++)
        if (dp[v] <= W)
            ans = v;

    cout << ans << '\n';
}
```

F LCS

atcoder/lcs.cpp

```
#include <bits/stdc++.h>
using namespace std;
using ll = long long;

int main() {
    cin.tie(0)->sync_with_stdio(0);

    string s, t;
    cin >> s >> t;
```

```
    int n = s.size(), m = t.size();

    vector<vector<int>> dp(n + 1, vector<int>(m + 1));

    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= m; j++) {
            if (s[i - 1] == t[j - 1])
                dp[i][j] = dp[i - 1][j - 1] + 1;
            else
                dp[i][j] = max(dp[i - 1][j], dp[i][j - 1]);
        }
    }

    string ans;

    int i = n, j = m;

    while (i && j) {
        if (s[i - 1] == t[j - 1]) {
            ans += s[i - 1];
            i--;
            j--;
        } else if (dp[i - 1][j] > dp[i][j - 1]) {
            i--;
        } else {
            j--;
        }
    }

    reverse(ans.begin(), ans.end());
}
```

G Longest Path

atcoder/longest_path.cpp

```
#include <bits/stdc++.h>
using namespace std;
using ll = long long;

int main() {
    cin.tie(0)->sync_with_stdio(0);

    int n, m;
    cin >> n >> m;

    vector<vector<int>> g(n);
    vector<int> in(n);

    while (m--) {
        int a, b;
        cin >> a >> b;
        --a, --b;
        g[a].push_back(b);
        in[b]++;
    }

    queue<int> q;
    for (int i = 0; i < n; i++)
        if (in[i] == 0)
            q.push(i);

    vector<int> dp(n);

    while (!q.empty()) {
        int v = q.front();
        q.pop();

        for (int u : g[v]) {
            dp[u] = max(dp[u], dp[v] + 1);

            if (--in[u] == 0)
                q.push(u);
        }
    }

    cout << *max_element(dp.begin(), dp.end()) << '\n';
}
```

H Grid 1

atcoder/grid1.cpp

```
#include <bits/stdc++.h>
using namespace std;
using ll = long long;

const int MOD = 1e9 + 7;

int main() {
    cin.tie(0)->sync_with_stdio(0);

    int h, w;
    cin >> h >> w;

    vector<string> a(h);
    for (string &s : a)
        cin >> s;

    vector<int> dp(w);
    dp[0] = 1;

    for (int i = 0; i < h; i++) {
        for (int j = 0; j < w; j++) {
            if (a[i][j] == '#') {
                dp[j] = 0;
                continue;
            }

            if (j)
                dp[j] = (dp[j] + dp[j - 1]) % MOD;
        }
    }

    cout << dp[w - 1] << '\n';
}
```

I Coins atcoder/coins.cpp

```
#include <bits/stdc++.h>
using namespace std;
using ll = long long;

int main() {
    cin.tie(0)->sync_with_stdio(0);

    int n;
    cin >> n;

    vector<double> dp(n + 1);
    dp[0] = 1;

    for (int i = 0; i < n; i++) {
        double p;
        cin >> p;

        for (int j = i + 1; j >= 0; j--) {
            dp[j] *= 1 - p;

            if (j)
                dp[j] += dp[j - 1] * p;
        }
    }

    double ans = 0;

    for (int heads = n / 2 + 1; heads <= n; heads++)
        ans += dp[heads];

    cout << fixed << setprecision(10) << ans << '\n';
}
```

J Sushi atcoder/sushi.cpp

```
#include <bits/stdc++.h>
using namespace std;
using ll = long long;

int n;
double dp[301][301][301];

double solve(int a, int b, int c) {
    if (a + b + c == 0)
        return 0;
```

```
double &res = dp[a][b][c];
if (res != 0)
    return res;

res = n;

if (a)
    res += a * solve(a - 1, b, c);

if (b)
    res += b * solve(a + 1, b - 1, c);

if (c)
    res += c * solve(a, b + 1, c - 1);

return res /= a + b + c;
}

int main() {
    cin.tie(0)->sync_with_stdio(0);

    cin >> n;

    int cnt[4] = {};

    for (int i = 0; i < n; i++) {
        int x;
        cin >> x;
        cnt[x]++;
    }

    cout << fixed << setprecision(15) << solve(cnt[1], cnt[2], cnt[3]) << '\n';
}
```

K Stones atcoder/stones.cpp

```
#include <bits/stdc++.h>
using namespace std;
using ll = long long;

int main() {
    cin.tie(0)->sync_with_stdio(0);

    int n, k;
    cin >> n >> k;

    vector<int> a(n);
    for (int &x : a)
        cin >> x;

    vector<bool> win(k + 1);

    for (int stones = 1; stones <= k; stones++) {
        for (int x : a) {
            if (x > stones)
                break;

            if (!win[stones - x]) {
                win[stones] = true;
                break;
            }
        }
    }

    cout << (win[k] ? "First" : "Second") << '\n';
}
```

L Deque atcoder/deque.cpp

```
#include <bits/stdc++.h>
using namespace std;
using ll = long long;

int main() {
    cin.tie(0)->sync_with_stdio(0);

    int n;
    cin >> n;
```

```
vector<ll> a(n), dp(n);

for (ll &x : a)
    cin >> x;

dp = a;

for (int len = 2; len <= n; len++) {
    for (int l = 0; l + len <= n; l++) {
        int r = l + len - 1;

        dp[l] = max(a[l] - dp[l + 1], a[r] - dp[l]);
    }
}

cout << dp[0] << '\n';
}
```

M Candies atcoder/candies.cpp

```
#include <bits/stdc++.h>
using namespace std;
using ll = long long;

const int MOD = 1e9 + 7;

int main() {
    cin.tie(0)->sync_with_stdio(0);

    int n, k;
    cin >> n >> k;

    vector<int> a(n);
    for (int &x : a)
        cin >> x;

    vector<ll> dp(k + 1), ndp(k + 1), pref(k + 1);
    dp[0] = 1;

    for (int x : a) {
        pref[0] = dp[0];

        for (int j = 1; j <= k; j++)
            pref[j] = (pref[j - 1] + dp[j]) % MOD;

        for (int j = 0; j <= k; j++) {
            ndp[j] = pref[j];

            if (j - x - 1 >= 0)
                ndp[j] = (ndp[j] - pref[j - x - 1] + MOD) % MOD;
        }

        dp.swap(ndp);
    }

    cout << dp[k] << '\n';
}
```

N Slimes atcoder/slimes.cpp

```
#include <bits/stdc++.h>
using namespace std;
using ll = long long;

int main() {
    cin.tie(0)->sync_with_stdio(0);

    int n;
    cin >> n;

    vector<ll> a(n), pref(n + 1);

    for (int i = 0; i < n; i++) {
        cin >> a[i];
        pref[i + 1] = pref[i] + a[i];
    }

    const ll INF = 4e18;
    vector<vector<ll>>> dp(n + 1, vector<ll>(n + 1));
```

```

for (int len = 2; len <= n; len++) {
    for (int l = 0; l + len <= n; l++) {
        int r = l + len;
        dp[l][r] = INF;

        for (int m = l + 1; m < r; m++)
            dp[l][r] = min(dp[l][r], dp[l][m] + dp[m][r] + pref[r] - pref[l]);
    }
}

cout << dp[0][n] << '\n';
}

```

O Matching atcoder/matching.cpp

```

#include <bits/stdc++.h>
using namespace std;
using ll = long long;

```

```

const int MOD = 1e9 + 7;

```

```

int main() {
    cin.tie(0)->sync_with_stdio(0);

```

```

    int n;
    cin >> n;

```

```

    vector<vector<int>> a(n, vector<int>(n));

```

```

    for (auto &row : a)
        for (int &x : row)
            cin >> x;

```

```

    vector<ll> dp(1 << n);
    dp[0] = 1;

```

```

    for (int mask = 0; mask < (1 << n); mask++) {
        int i = __builtin_popcount(mask);

```

```

        if (i == n)
            continue;

```

```

        for (int j = 0; j < n; j++) {
            if (a[i][j] && !(mask >> j & 1)) {
                int nxt = mask | (1 << j);
                dp[nxt] = (dp[nxt] + dp[mask]) % MOD;
            }
        }
    }
}

```

```

cout << dp[(1 << n) - 1] << '\n';
}

```

P Independent Set atcoder/independent-set.cpp

```

#include <bits/stdc++.h>
using namespace std;
using ll = long long;

```

```

const ll MOD = 1e9 + 7;

```

```

vector<vector<int>> g;

```

```

pair<ll, ll> dfs(int v, int p) {
    ll white = 1, black = 1;

```

```

    for (int u : g[v]) {
        if (u == p)
            continue;

```

```

        auto [w, b] = dfs(u, v);

```

```

        white = white * (w + b) % MOD;
        black = black * w % MOD;
    }

```

```

    return {white, black};
}

```

```

int main() {

```

```

    cin.tie(0)->sync_with_stdio(0);

```

```

    int n;
    cin >> n;

```

```

    g.resize(n);

```

```

    for (int i = 0; i < n - 1; i++) {
        int a, b;
        cin >> a >> b;
        --a, --b;

```

```

        g[a].push_back(b);
        g[b].push_back(a);
    }

```

```

    auto [white, black] = dfs(0, -1);

```

```

    cout << (white + black) % MOD << '\n';
}

```

Q Flowers atcoder/flowers.cpp

```

#include <bits/stdc++.h>
using namespace std;
using ll = long long;

```

```

struct Fenwick {
    int n;
    vector<ll> bit;

```

```

    Fenwick(int n) : n(n), bit(n + 1) {}

```

```

    void update(int i, ll x) {
        for (; i <= n; i += i & -i)
            bit[i] = max(bit[i], x);
    }

```

```

    ll query(int i) {
        ll res = 0;

```

```

        for (; i > 0; i -= i & -i)
            res = max(res, bit[i]);

```

```

        return res;
    }
};

```

```

int main() {
    cin.tie(0)->sync_with_stdio(0);

```

```

    int n;
    cin >> n;

```

```

    vector<int> h(n);
    vector<ll> a(n);

```

```

    for (int &x : h)
        cin >> x;
    for (ll &x : a)
        cin >> x;

```

```

    Fenwick fw(n);

```

```

    for (int i = 0; i < n; i++) {
        ll cur = fw.query(h[i] - 1) + a[i];
        fw.update(h[i], cur);
    }

```

```

    cout << fw.query(n) << '\n';
}

```

R Walk atcoder/walk.cpp

```

#include <bits/stdc++.h>
using namespace std;
using ll = long long;

```

```

const ll MOD = 1e9 + 7;

```

```

using Matrix = vector<vector<ll>>;

```

```

Matrix mul(const Matrix &a, const Matrix &b) {
    int n = a.size();
    Matrix c(n, vector<ll>(n));

```

```

    for (int i = 0; i < n; i++)
        for (int k = 0; k < n; k++)
            if (a[i][k])
                for (int j = 0; j < n; j++)
                    c[i][j] = (c[i][j] + a[i][k] * b[k][j]) % MOD;

```

```

    return c;
}

```

```

int main() {
    cin.tie(0)->sync_with_stdio(0);

```

```

    int n;
    ll k;
    cin >> n >> k;

```

```

    Matrix a(n, vector<ll>(n));
    Matrix res(n, vector<ll>(n));

```

```

    for (auto &row : a)
        for (ll &x : row)
            cin >> x;

```

```

    for (int i = 0; i < n; i++)
        res[i][i] = 1;

```

```

    while (k) {
        if (k & 1)
            res = mul(res, a);

```

```

        a = mul(a, a);
        k >>= 1;
    }

```

```

    ll ans = 0;

```

```

    for (auto &row : res)
        for (ll x : row)
            ans = (ans + x) % MOD;

```

```

    cout << ans << '\n';
}

```

S Digit Sum atcoder/digit-sum.cpp

```

#include <bits/stdc++.h>
using namespace std;
using ll = long long;

```

```

const int MOD = 1e9 + 7;

```

```

int main() {
    cin.tie(0)->sync_with_stdio(0);

```

```

    string k;
    int d;
    cin >> k >> d;

```

```

    vector<vector<int>> dp(d, vector<int>(2));
    dp[0][0] = 1;

```

```

    for (char c : k) {
        int lim = c - '0';
        vector<vector<int>> ndp(d, vector<int>(2));

```

```

        for (int r = 0; r < d; r++) {
            for (int less = 0; less < 2; less++) {
                int hi = less ? 9 : lim;

```

```

                for (int x = 0; x <= hi; x++) {
                    int nr = (r + x) % d;
                    int nl = less || x < lim;

```

```

                    ndp[nr][nl] += dp[r][less];

```

```

        if (ndp[nr][nl] >= MOD)
            ndp[nr][nl] -= MOD;
    }
}

dp.swap(ndp);
}

int ans = (dp[0][0] + dp[0][1] - 1 + MOD) % MOD;

cout << ans << '\n';
}

```

T Permutation

[atcoder/permutation.cpp](#)

```

#include <bits/stdc++.h>
using namespace std;
using ll = long long;

const int MOD = 1e9 + 7;

int main() {
    cin.tie(0) -> sync_with_stdio(0);

    int n;
    string s;
    cin >> n >> s;

    vector<ll> dp(n), ndp(n), pref(n + 1);
    dp[0] = 1;

    for (int i = 1; i < n; i++) {
        pref[0] = 0;

        for (int j = 0; j < i; j++)
            pref[j + 1] = (pref[j] + dp[j]) % MOD;

        fill(ndp.begin(), ndp.end(), 0);

        for (int j = 0; j <= i; j++) {
            if (s[i - 1] == '<' ) {
                ndp[j] = pref[j];
            } else {
                ndp[j] = (pref[i] - pref[j] + MOD) % MOD;
            }
        }

        dp.swap(ndp);
    }

    ll ans = 0;

    for (int j = 0; j < n; j++)
        ans = (ans + dp[j]) % MOD;

    cout << ans << '\n';
}

```

U Grouping

[atcoder/grouping.cpp](#)

```

#include <bits/stdc++.h>
using namespace std;
using ll = long long;

int main() {
    cin.tie(0) -> sync_with_stdio(0);

    int n;
    cin >> n;

    vector<vector<ll>> a(n, vector<ll>(n));

    for (auto &row : a)
        for (ll &x : row)
            cin >> x;

    int N = 1 << n;

    vector<ll> score(N);

```

```

    for (int mask = 1; mask < N; mask++) {
        int i = __builtin_ctz(mask);
        int rest = mask ^ (1 << i);

        score[mask] = score[rest];

        for (int j = 0; j < n; j++)
            if (rest >> j & 1)
                score[mask] += a[i][j];
    }

    vector<ll> dp(N);
    dp[0] = 0;

    for (int mask = 1; mask < N; mask++) {
        int bit = mask & -mask;
        dp[mask] = score[mask];

        for (int sub = mask; sub; sub = (sub - 1) & mask) {
            if (!(sub & bit))
                continue;

            dp[mask] = max(dp[mask], score[sub] + dp[mask ^ sub]);
        }
    }

    cout << dp[N - 1] << '\n';
}

```

V Subtree

[atcoder/subtree.cpp](#)

```

#include <bits/stdc++.h>
using namespace std;
using ll = long long;

int main() {
    cin.tie(0) -> sync_with_stdio(0);

    int n;
    ll mod;
    cin >> n >> mod;

    vector<vector<int>> g(n);

    for (int i = 0; i < n - 1; i++) {
        int a, b;
        cin >> a >> b;
        --a, --b;
        g[a].push_back(b);
        g[b].push_back(a);
    }

    vector<int> par(n, -1), order = {0};
    par[0] = 0;

    for (int i = 0; i < n; i++) {
        int v = order[i];

        for (int u : g[v]) {
            if (u == par[v])
                continue;
            par[u] = v;
            order.push_back(u);
        }
    }

    vector<ll> down(n, 1), up(n, 1), ans(n);

    for (int i = n - 1; i >= 0; i--) {
        int v = order[i];

        for (int u : g[v]) {
            if (par[u] == v)
                down[v] = down[v] * (down[u] + 1) % mod;
        }
    }

    for (int v : order) {
        vector<int> ch;

```

```

        for (int u : g[v])
            if (par[u] == v)
                ch.push_back(u);

        int m = ch.size();
        vector<ll> pref(m + 1, 1), suf(m + 1, 1);

        for (int i = 0; i < m; i++)
            pref[i + 1] = pref[i] * (down[ch[i]] + 1) % mod;

        for (int i = m - 1; i >= 0; i--)
            suf[i] = suf[i + 1] * (down[ch[i]] + 1) % mod;

        ans[v] = down[v] * up[v] % mod;

        for (int i = 0; i < m; i++) {
            ll other = pref[i] * suf[i + 1] % mod;
            up[ch[i]] = (1 + up[v] * other) % mod;
        }

        for (ll x : ans)
            cout << x << '\n';
    }
}

```

W Intervals

[atcoder/intervals.cpp](#)

```

#include <bits/stdc++.h>
using namespace std;
using ll = long long;

struct SegTree {
    int n;
    vector<ll> st, lazy;

    SegTree(int n) : n(n), st(4 * n), lazy(4 * n) {}

    void push(int p) {
        if (!lazy[p])
            return;

        for (int q : {p * 2, p * 2 + 1}) {
            st[q] += lazy[p];
            lazy[q] += lazy[p];
        }

        lazy[p] = 0;
    }

    void add(int p, int l, int r, int ql, int qr, ll x) {
        if (qr < l || r < ql)
            return;

        if (ql <= l && r <= qr) {
            st[p] += x;
            lazy[p] += x;
            return;
        }

        push(p);

        int m = (l + r) / 2;
        add(p * 2, l, m, ql, qr, x);
        add(p * 2 + 1, m + 1, r, ql, qr, x);

        st[p] = max(st[p * 2], st[p * 2 + 1]);
    }

    void setval(int p, int l, int r, int pos, ll x) {
        if (l == r) {
            st[p] = x;
            lazy[p] = 0;
            return;
        }

        push(p);

        int m = (l + r) / 2;

```

```

    if (pos <= m)
        setval(p * 2, l, m, pos, x);
    else
        setval(p * 2 + 1, m + 1, r, pos, x);

    st[p] = max(st[p * 2], st[p * 2 + 1]);
}

void add(int l, int r, ll x) { add(1, 0, n - 1, l, r, x); }

void setval(int pos, ll x) { setval(1, 0, n - 1, pos, x); }

ll best() { return st[1]; }
};

```

```

int main() {
    cin.tie(0)->sync_with_stdio(0);

    int n, m;
    cin >> n >> m;

    vector<vector<pair<int, ll>>> ending(n + 1);

```

```

    while (m--) {
        int l, r;
        ll a;
        cin >> l >> r >> a;
        ending[r].push_back({l, a});
    }

```

```

SegTree seg(n + 1);

```

```

for (int r = 1; r <= n; r++) {
    seg.setval(r, seg.best());

    for (auto [l, a] : ending[r])
        seg.add(l, r, a);
}

```

```

    cout << seg.best() << '\n';
}

```

X Tower atcoder/tower.cpp

```

#include <bits/stdc++.h>
using namespace std;
using ll = long long;

```

```

struct Block {
    int w, s;
    ll v;
};

```

```

int main() {
    cin.tie(0)->sync_with_stdio(0);

```

```

    int n;
    cin >> n;

```

```

    vector<Block> a(n);
    int maxw = 0;

```

```

    for (auto &[w, s, v] : a) {
        cin >> w >> s >> v;
        maxw = max(maxw, w + s);
    }

```

```

    sort(a.begin(), a.end(),
         [](const Block &a, const Block &b) { return a.w + a.s < b.w + b.s; });

```

```

    vector<ll> dp(maxw + 1, -1);
    dp[0] = 0;

```

```

    for (auto [w, s, v] : a) {
        for (int cur = s; cur >= 0; cur--) {
            if (dp[cur] == -1)
                continue;

            dp[cur + w] = max(dp[cur + w], dp[cur] + v);
        }
    }

```

```

    }

    cout << *max_element(dp.begin(), dp.end()) << '\n';
}

```

Y Grid 2 atcoder/grid2.cpp

```

#include <bits/stdc++.h>
using namespace std;
using ll = long long;

```

```

const ll MOD = 1e9 + 7;

```

```

ll modpow(ll a, ll e) {
    ll r = 1;

```

```

    while (e) {
        if (e & 1)
            r = r * a % MOD;
        a = a * a % MOD;
        e >>= 1;
    }

```

```

    return r;
}

```

```

int main() {
    cin.tie(0)->sync_with_stdio(0);

```

```

    int h, w, n;
    cin >> h >> w >> n;

```

```

    vector<pair<int, int>> p(n);

```

```

    for (auto &[r, c] : p)
        cin >> r >> c;

```

```

    p.push_back({h, w});
    sort(p.begin(), p.end());

```

```

    int mx = h + w;
    vector<ll> fact(mx + 1), invfact(mx + 1);

```

```

    fact[0] = 1;
    for (int i = 1; i <= mx; i++)
        fact[i] = fact[i - 1] * i % MOD;

```

```

    invfact[mx] = modpow(fact[mx], MOD - 2);

```

```

    for (int i = mx; i >= 1; i--)
        invfact[i - 1] = invfact[i] * i % MOD;

```

```

    auto C = [&](int n, int k) -> ll {
        if (k < 0 || k > n)
            return 0;
        return fact[n] * invfact[k] % MOD * invfact[n - k] % MOD;
    };

```

```

    vector<ll> dp(n + 1);

```

```

    for (int i = 0; i <= n; i++) {
        auto [r, c] = p[i];

```

```

        dp[i] = C(r + c - 2, r - 1);

```

```

        for (int j = 0; j < i; j++) {
            auto [pr, pc] = p[j];

```

```

            if (pc > c)
                continue;

```

```

            int dr = r - pr;
            int dc = c - pc;

```

```

            dp[i] = (dp[i] - dp[j] * C(dr + dc, dr)) % MOD;
        }

```

```

        if (dp[i] < 0)
            dp[i] += MOD;
    }
}

```

```

    cout << dp[n] << '\n';
}

```

Z Frog 3 atcoder/frog3.cpp

```

#include <bits/stdc++.h>
using namespace std;
using ll = long long;

```

```

struct Line {
    ll m, b;

    ll get(ll x) { return m * x + b; }
};

```

```

bool bad(Line a, Line b, Line c) {
    return (__int128)(b.b - a.b) * (b.m - c.m) >=
        (__int128)(c.b - b.b) * (a.m - b.m);
}

```

```

int main() {
    cin.tie(0)->sync_with_stdio(0);

```

```

    int n;
    ll C;
    cin >> n >> C;

```

```

    vector<ll> h(n), dp(n);
    for (ll &x : h)
        cin >> x;

```

```

    deque<Line> hull;

```

```

    hull.push_back({-2 * h[0], h[0] * h[0]});

```

```

    for (int i = 1; i < n; i++) {
        ll x = h[i];

```

```

        while (hull.size() >= 2 && hull[0].get(x) >= hull[1].get(x))
            hull.pop_front();

```

```

        dp[i] = x * x + C + hull[0].get(x);

```

```

        Line cur = {-2 * x, dp[i] + x * x};

```

```

        while (hull.size() >= 2 && bad(hull[hull.size() - 2], hull.back(), cur))
            hull.pop_back();

```

```

        hull.push_back(cur);
    }

```

```

    cout << dp[n - 1] << '\n';
}

```